(a) Aspartame is as sweet as table sugar, but unlike table sugar, it is not readily oxidized in human body due to lack of requisite enzymes.
(b) When aspartame is used in food processing , the sweet taste remains, but it becomes resistant to oxidation.
(c) Aspartame is as sweet as sugar, but after ingestion into the body, it is converted into metabolites that yield no calories
(d) Aspartame is several times sweeter than table sugar, hence food items made with small quantities of aspartame yield fewer calories on oxidation.

Ans. (d) : Aspartame is an artificial sweetener sold in the market. It is many times sweeter than normal sugar or sucrose. Therefore, a very small amount of aspartame is sufficient to produce a sweet taste. Therefore, its calorie fraction is negligible.

• Botany

70. Regular intake of fresh fruits and vegetables is recommended in the diet since they are a good source of antioxidants. How do antioxidants help a person maintain health and promote longevity?
(a) They activate the enzymes necessary for vitamins synthesis in the body and help prevent vitamin deficiency.
(b) They prevent excessive oxidation of carbohydrates, fats and proteins in the body and help avoid unnecessary wastage of energy.
(c) They neutralize the free radicals produced in the body during metabolism.
(d) They activate certain genes in the cells of the body and help delay and ageing process.

Ans. (c) : It is desirable to take fresh fruits and vegetables in the diet regularly as they are good source of anti-oxidant element. Anti-oxidant elements are helpful in keeping a person healthy and maintain longevity as they neutralize the free radicals produced in the body as a byproduct of metabolism. Oxidation is a chemical reaction in which electrons transfer from a substance to an oxidizing agent. Oxidation reactions can initiate chain reactions that damage cells.

71. When the bark of a tree is removed in a circular fashion all around near its base, it gradually dries up and dies because–
(a) Water from soil cannot rise to aerial parts
(b) Roots are starved of energy
(c) Tree is infected by soil microbes.
(d) Roots do not receive oxygen for respiration.

Ans. (b): Two types of tissues are found in vascular plants: phloem and xylem. Phloem is the innermost layer of the bark of the tree. Its main function is to transport soluble organic matter produced in the process of photosynthesis. Phloem will also be damaged by removing the bark of a tree from all four sides near the base of the tree, due to which the nutrients produced by the leaves in the process of photosynthesis will not reach the roots, the roots will be deprived of energy, and the tree will gradually dry and slowly will be dead.

• Zoology

72. A married couple adopted a male child. A few years later, twin boys were born to them. The blood group of the couple is AB positive and O negative. The blood group of the three sons is A positive, B positive, and O positive. The blood group of the adopted son is –
(a) O positive
(b) A positive
(c) B positive
(d) Cannot be determined on the basis of the given data.

Ans. (a): The blood group of the parents are AB and O; the child's blood group may be A or B and since the body is adopted and it is given in the question that "the blood group of the three sons are A^+, B^+, O^+". The correct answer is O^+.

• Agriculture Animal Husbandry & Dairy

73. At present, scientists can determine the arrangement or relative positions of genes or DNA sequences on a chromosome. How does this knowledge benefit us?
1. It is possible to know the pedigree of livestock.
2. It is possible to understand the causes of all human diseases.
3. It is possible to develop diseases-resistant animal breeds.
Which of the statements given above is/are correct?
(a) 1 and 2 only (b) 2 only
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c) : The DNA sequence help in understanding the genealogies of the animals. In addition, various genetic diseases such as Alzheimer's, Myotonic dystrophy, cystic fibrosis and inability of gene and defects arising in them can be cured by finding out the cause of many diseases.

• Computers

74. What is the difference between Bluetooth and Wi-Fi Devices?
(a) Bluetooth uses 2.4 GHz radio frequency band, whereas Wi-Fi can use 2.4 GHz or 5 GHz frequency band.
(b) Bluetooth is used for Wireless Local Area Networks (WLAN) only, where Wi-Fi is used for Wireless Wide Area Networks (WWAN) only.
(c) When information is transmitted between two devices using Bluetooth technology, the devices have to be in a line of sight of each other, but when Wi-Fi technology is used the devices need not be in the line of sight of each other
(d) The Statements (a) and (b) given above are correct in this context

Ans. (a): Bluetooth uses a radio frequency spectrum in the range of 2.402 GHz to 2.480 GHz and Wi-Fi is IEEE 802.11 standard rate based technology using 2.4 GHz, 35 GHz and 5 GHz frequency spectrum.

75. A new optical disc format known as the Blu-ray Disc (BD) is becoming popular. In what way is it different from the traditional DVD ?
1. DVD supports Standard Definition video while BD supports High Definition video.
2. Compared to a DVD, the BD format has several times more storage capacity.
3. Thickness of BD is 2.4mm while that of DVD is 1.2 mm.
Which of the statements given above is/are correct?
(a) 1 only (b) 1 and 2 only
(c) 2 and only (d) 1, 2 and 3

Ans. (b) : The Blu-ray disc is the next- generation optical disc dedicated to high definition video. DVD discs can store a maximum of 4.7 GB of data and a single-layer Blu-ray disc can store 25 GB of data on an average.

• Science & Technology

76. Satellites used for telecommunication relay are kept in a geostationary orbit. A satellite is said to be in such an orbit when:
1. The orbit is geosynchronous.
2. The orbit is circular
3. The orbit lies in the plane of the Earth's equator.
4. The orbit is at an altitude of 22,236 km.
Select the correct answer using the codes given below :
(a) 1, 2 and 3 only (b) 1, 3 and 4 only
(c) 2 and 4 only (d) 1, 2, 3, and 4

Ans. (a) : A geostationary orbit is a circular geosynchronous orbit in which the satellite is placed at an altitude of approximately 35786 km above mean sea level. The orbit lies in the plane of the earth's equator. An object in such an orbit that has an orbital period equal to earth's rotational period and thus appear motionless.

77. Microbial fuel cells are considered a source of sustainable energy. Why?
1. They use living organisms as catalysts to generate electricity from certain substrates.
2. They use a variety of inorganic materials as substrates.
3. They can be installed in waste water treatment plants to cleanse water and produce electricity.
Which of the statement given above is/are correct?
(a) 1only (b) 2 and 3
(c) 1 and 3 only (d) 1, 2 and 3

Ans. (c): Microbial fuel cell (MFC) converts the chemical energy available in a biodegradable substrate directly into electrical energy. MFC is also considered an attractive alternative to power generation because it contains organic matter such as sucrose, glucose, acetate, starch format, methanol, protein etc. are used as substrates. Also, some inorganic substances such as sulfides are also used as substrates.

78. The formation of ozone hole in the Antarctic region has been a cause of concern. What could be the reason for a formation of this hole
(a) Presence of prominent troposphere turbulence, and inflow of chlorofluorocarbons.
(b) Presence of prominent polar front and stratospheric clouds, and inflow of chlorofluorocarbons.
(c) Absence of polar front and stratospheric clouds; and inflow of methane and chlorofluorocarbons.
(d) Increased temperature at polar region due to global warming.

Ans. (b) : The formation of an ozone hole in the Antarctic region is a matter of concern, the possible reason for the formation of this hole in the presence of specific polar atmospheres and stratospheric clouds and the inflow of chlorofluorocarbons. Nitric acid present in polar stratospheric clouds reacts with chlorofluorocarbons to form chlorine which is responsible for the photochemical destruction of the ozone layer.